Compressed BundleFuse: Inference Algorithm

1 Conventions

Indices: items $i, c \in \{1, \dots, n\}$; bundles $b \in \{1, \dots, m\}$; seeds $s \in S$. For a matrix M, $M_{:,J}$ is the sub-matrix of columns in J, and $M_{j,:}$ is row j. $\odot$ is the Hadamard (element-wise) product. $(x)_+ = \max(x, 0)$. For a vector $\boldsymbol{r} \in \mathbb{R}^n$, its *column standardisation* is

$$\mathrm{z}(\boldsymbol{r}) \in \mathbb{R}^n, \qquad \mathrm{z}(\boldsymbol{r})_c = \frac{r_c - \mu(\boldsymbol{r})}{\sigma(\boldsymbol{r}) + \varepsilon},$$

where $\mu(\boldsymbol{r}), \sigma(\boldsymbol{r})$ are the mean and (population) standard deviation of the n entries of $\boldsymbol{r}$, and $\varepsilon = 10^{-8}$.

2 Inputs

The algorithm is fully specified by the following inputs; all are fixed offline except the query S.

Query.

- $n \in \mathbb{N}$: number of catalogue items.
- $S \subseteq \{1, \dots, n\}$, $S \neq \varnothing$: the seed set (partial bundle).
- $K \in \mathbb{N}$, $1 \leq K \leq n - |S|$: number of items to return.

Collaborative artifacts.

- $B \in \mathbb{R}^{m \times n}$: bundle incidence ($B_{b,i}$ = weight of item i in bundle b); $B^{\mathrm{bin}} \in \{0, 1\}^{m \times n}$: its binary pattern.
- $\boldsymbol{f} \in \mathbb{R}^n_{\geq 0}$: item frequency, $f_i = \sum_b B^{\mathrm{bin}}_{b,i}$.
- $\boldsymbol{f}^c \in \mathbb{R}^n_{>0}$: co-occurrence degree, $f^c_i = \left(\sum_{j \neq i} C_{ij}\right) + 10^{-9}$, where $C = B^\top B$ with zeroed diagonal (never materialised; see Step 3).
- $T \in \mathbb{R}_{>0}$: total co-occurrence mass, $T = \sum_i \sum_{j \neq i} C_{ij}$ (scalar).
- $\boldsymbol{a} \in \mathbb{R}^n$: item normaliser ($a_i = (f_i + 5)^{-\alpha_{\deg}}$); $\boldsymbol{v} \in \mathbb{R}^m$: bundle normaliser (volume weights); $\beta \in \mathbb{R}_{>0}$: clique coefficient.
- *(optional, auxiliary co-occurrence, e.g. household)* $X_w \in \mathbb{R}^{u \times n}$: degree-weighted user/basket incidence over u contexts.

Embedding and taxonomy artifacts.

- $U \in \mathbb{R}^{n \times d}$: collaborative SVD fingerprint (d latent dims).
- $\{E^{(k)} \in \mathbb{R}^{n \times r_k}\}_{k \in \mathcal{K}_{\text{cont}}}$: rank-$r_k$ content embeddings, one per content signal k in the content index set $\mathcal{K}_{\text{cont}}$.
- $\boldsymbol{\ell} \in \{1, \ldots, N_{\text{cat}}\}^n$: item category labels; $P \in \mathbb{R}^{N_{\text{cat}} \times N_{\text{cat}}}$: category co-occurrence.

Fitted scalars (selected on validation).

- Signal index set $\mathcal{G}$ (all signals with nonzero weight) and weights $\{w_k \geq 0\}_{k \in \mathcal{G}}$.
- Prior subset $\mathcal{P} \subseteq \mathcal{G}$ with weights $\{w^p_k \geq 0\}_{k \in \mathcal{P}}$ and scale $s \in \mathbb{R}_{>0}$.
- Shrinkage rate $\kappa \in \mathbb{R}_{>0}$.

3 Algorithm

4 Guarantees and cost

- **Well-defined.** Every quantity used in a step is an input (Section 2) or defined in an earlier step; all dimensions match as annotated. A signal $k \notin \mathcal{G}$ is never computed.
- **Exactness.** Steps 5, 8–10, 13 reproduce the materialised-matrix scores exactly: $B^\top(B\mathbf{1}) = C\mathbf{1}$, and $G_{c,s} = C_{c,s}$ for candidates $c \notin S$ (the removed diagonal affects only $c \in S$, which are masked in Step 26). The sole approximation is the rank-r_k truncation of the content embeddings $E^{(k)}$ in Step 18.
- **Cost.** Time $O\big(|\mathcal{G}|\, n + \text{nnz}(B) + |S|\, \text{nnz}(B)\big)$ per query (the last term only when ppmi $\in \mathcal{G}$); the $O(n \log K)$ selection is dominated. Resident memory is the inputs; the per-query working set is $O(|\mathcal{G}|\, n)$ for the score vectors plus $O(n\, |S|)$ for the slice G.

Algorithm 1 COMPLETE(S) — score all n candidates and return the top K

Require: all inputs of Section 2.
Ensure: an ordered list of K item indices.

$\mathbf{1} \leftarrow \mathbf{0} \in \mathbb{R}^n$; $\mathbf{1}_i \leftarrow 1$ for each $i \in S$ ▷ seed indicator, $\mathbf{1} \in \mathbb{R}^n$
$\mathbf{1}^w \leftarrow \boldsymbol{a} \odot \mathbf{1} \in \mathbb{R}^n$ ▷ item-normalised indicator
initialise the score table $\{\boldsymbol{z}_k\}_{k \in \mathcal{G}}$, each $\boldsymbol{z}_k \in \mathbb{R}^n$
Collaborative signals — recomputed from B (Steps 4–7); compute only those $k \in \mathcal{G}$.
if cooc $\in \mathcal{G}$ **then**
  $\boldsymbol{z}_{\text{cooc}} \leftarrow \mathrm{z}\big(B^\top (B\,\mathbf{1})\big)$ ▷ $B\mathbf{1} \in \mathbb{R}^m$, then $B^\top(\cdot) \in \mathbb{R}^n$; equals $\mathrm{z}(C\mathbf{1})$
end if
if ppmi $\in \mathcal{G}$ **then**
  $G \leftarrow B^\top B_{:,S} \in \mathbb{R}^{n \times |S|}$ ▷ $G_{c,s} = C_{c,s}$ for $c \notin S$
  $(\boldsymbol{r})_c \leftarrow \sum_{s \in S} \big(\log \frac{G_{c,s}\, T}{f_c^c\, f_s^c}\big)_+ \cdot \mathbb{1}[G_{c,s} > 0]$, for all c
  $\boldsymbol{z}_{\text{ppmi}} \leftarrow \mathrm{z}(\boldsymbol{r})$
end if
if simp $\in \mathcal{G}$ **then**
  $\boldsymbol{o} \leftarrow \boldsymbol{v} \odot (B\,\mathbf{1}^w) \in \mathbb{R}^m$; $\boldsymbol{q} \leftarrow B^{\text{bin}}\,\mathbf{1} \in \mathbb{R}^m$ ▷ per-bundle overlap; # seeds in bundle
  $\boldsymbol{z}_{\text{simp}} \leftarrow \mathrm{z}\Big(\underbrace{B^\top \boldsymbol{o}}_{\text{pairwise}} + \underbrace{B^\top\big(\boldsymbol{o} \odot \beta\,(\boldsymbol{q} - \mathbf{1})_+\big)}_{\text{incidence surplus}} \Big)$
end if
if aux $\in \mathcal{G}$ **then** ▷ optional, if X_w provided
  $\boldsymbol{z}_{\text{aux}} \leftarrow \mathrm{z}\big(X_w^\top (X_w\,\mathbf{1}^w)\big)$
end if
Embedding signals — dot products in stored tables.
if csvd $\in \mathcal{G}$ **then**
  $\boldsymbol{u}_S \leftarrow \sum_{s \in S} U_{s,:} \in \mathbb{R}^d$; $\boldsymbol{z}_{\text{csvd}} \leftarrow \mathrm{z}\big(U\,\boldsymbol{u}_S\big)$
end if
for all $k \in \mathcal{G} \cap \mathcal{K}_{\text{cont}}$ **do**
  $\boldsymbol{e}_S \leftarrow \frac{1}{|S|} \sum_{s \in S} E^{(k)}_{s,:} \in \mathbb{R}^{r_k}$; $\boldsymbol{z}_k \leftarrow \mathrm{z}\big(E^{(k)}\,\boldsymbol{e}_S\big)$
end for
Cheap signals.
if pop $\in \mathcal{G}$ **then** $\boldsymbol{z}_{\text{pop}} \leftarrow \mathrm{z}\big(\log(\mathbf{1} + \boldsymbol{f})\big)$
end if
if cat $\in \mathcal{G}$ **then**
  $(\boldsymbol{r})_c \leftarrow \sum_{s \in S} P_{\ell_c, \ell_s}$ for all c; $\boldsymbol{z}_{\text{cat}} \leftarrow \mathrm{z}(\boldsymbol{r})$
end if
Fuse, frequency-shrink, rank.
$\boldsymbol{Y} \leftarrow \sum_{k \in \mathcal{G}} w_k\, \boldsymbol{z}_k \in \mathbb{R}^n$
$\boldsymbol{\mu} \leftarrow s \sum_{k \in \mathcal{P}} w_k^p\, \boldsymbol{z}_k \in \mathbb{R}^n$ ▷ cold-start prior; $\mathcal{P} \subseteq \mathcal{G}$
for $c = 1, \ldots, n$ **do**
  $u_c \leftarrow \frac{\kappa}{f_c + \kappa}$; $\theta_c \leftarrow (1 - u_c)\, Y_c + u_c\, \mu_c$
end for
$\theta_s \leftarrow -\infty$ for all $s \in S$ ▷ seeds are not candidates
return indices of the K largest entries of $\boldsymbol{\theta}$, in decreasing order (ties broken by smaller index)